



João Francisco de Veloso e Viana

Av. Dom Dinis, 68 9ºB, 2675-328
Odivelas, Lisboa, Portugal
+351 969 823 979
jfvvchico@hotmail.com

PERSONAL DATA

Date of Birth : 13/05/1998
Nationality : Portuguese
Languages: Portuguese, English
Occupation: Student

EDUCATION

- 5 Years of Trumpet, Conservatório de Música e Artes do Dão (2008-2013)
- Secondary Degree in Sciences and Technologies, Escola Secundária Alves Martins (2013-2016)
- Bachelor's Degree in Physics, Faculdade de Ciências da Universidade de Lisboa (2016-2019)
- Master's Student in Nuclear and Particle Physics, Faculdade de Ciências da Universidade de Lisboa (2019-2022)
- PhD Student in Particle Physics - "The early universe as a window to particle physics", Faculdade de Ciências da Universidade de Lisboa (since 2022)

ON GOING PROJECTS

- Calculating the spectrum of primordial gravitational waves produced in a first order electroweak phase transition.
- Calculate the baryon asymmetry of the Universe produced by a strong first order electroweak phase transition.

AWARDS

- FCIências.ID – Menção Honrosa (2016/2017) - Faculdade de Ciências da Universidade de Lisboa
- Prémio de Mérito Académico (2016/2017) - Faculdade de Ciências da Universidade de Lisboa
- Prémio de Mérito Académico (2017/2018) - Faculdade de Ciências da Universidade de Lisboa
- Prémio de Mérito Académico (2018/2019) - Faculdade de Ciências da Universidade de Lisboa

EXPERIENCE

- 6 Months Scientific Initiation Fellowships with : Dr. Mykola Tasinkevych on "Engineering controlled motion of chemically active microswimmers" at C.F.T.C. , Ref. IF/00322/2015
- Master's Fellowships with : Dr. Rui Santos on "Standard Model Extensions at the LHC/Centro de Física Teórica e Computacional da Universidade de Lisboa" at C.F.T.C. , Ref. CERN/FIS-PAR/0014/2019
- PhD's Fellowships with : Dr. Rui Santos on "The early universe as a window to particle physics" at C.F.T.C. , Ref. PRT/BD/154191/2022

TALKS

- Harmonia Meeting VII - Electroweak corrections to the invisible Higgs decay in the Singlet Extension of the Standard Model
- Workshop on Multi-Higgs Models 2022 - Impact of SM parameters on the spectrum

of primordial gravitational waves

- BSM2 - Beyond the Standard Model BrainStorming Meeting - Electroweak Corrections to the CxSM Scalar Sector
- CFTC Summer Internships '23 - The sound of vacuum decay
- Probing the early Universe using BSMPT v3 Workshop on Multi Higgs models (2024)
- Probing the early Universe with BSMPT v3: CP in the Dark, a test case LHC Higgs Working Group WG3 (BSM) – Extended Higgs Sector subgroup meeting (2024)
- BSMPT v3 A Tool for Phase Transitions and Primordial Gravitational Waves in Extended Higgs Sectors Asymptotic Safety meets Particle Physics and Friends (2024)

- PUBLICATIONS**
- Controlling the pandemic during the SARS-CoV-2 vaccination rollout - Viana, J., van Dorp, C.H., Nunes, A. et al. - Nat Commun 12, 3674 (2021)
 - Impact of SM parameters and of the vacua of the Higgs potential in gravitational waves detection. - Felipe F. Freitas, Gabriel Lourenço, António P. Morais, André Nunes, João Olívia, Roman Pasechnik, Rui Santos, & João Viana. - JCAP03(2022)046
 - Gravitational-Wave Signatures of Chiral-Symmetric Technicolor - Hao Yang, Felipe F. Freitas, Antonino Marciano, António P. Morais, Roman Pasechnik, João Viana - PLB-137162(2022)
 - One-loop Corrections to the Higgs Boson Invisible Decay in a Complex Singlet Extension of the SM. - Felix Egle, Margarete Mühlleitner, Rui Santos, João Viana - PRD 106, 095030 (2022)
 - Gravitational echoes of lepton number symmetry breaking with light and ultralight Majorons - Andrea Addazi, Antonino Marcianò, António P. Morais, Roman Pasechnik, João Viana, Hao Yang - JCAP09(2023)026
 - Electroweak corrections to Higgs boson decays in a Complex Singlet extension of the SM and their phenomenological impact - Felix Egle, Margarete Mühlleitner, Rui Santosb, and João Viana - JCAP09(2023)026
 - BSMPT v3 A Tool for Phase Transitions and Primordial Gravitational Waves in Extended Higgs Sectors - Philipp Basler, Lisa Biermann, Margarete Mühlleitner, Jonas Müller, Rui Santos, João Viana - arXiv:2404.19037

- POSTERS**
- Equation of State of Active Colloidal Particles - J. Viana, M.Tasinkevych - 2nd Portuguese Condensed Matter Physics Meeting

- CONFERENCES**
- 2nd Portuguese Condensed Matter Physics Meeting
 - 69th Lindau Nobel Laureate Meeting
 - Harmonia Meeting VII
 - Workshop on Multi-Higgs Models 2022
 - 12th International IDPASC school and workshop - Granada 2023
 - BSM² - Beyond the Standard Model BrainStorming Meeting
 - ML4Physics@Ljubljana school

- EXPERIENCE
TEACHING**
- CFTC Summer Internship Program 2023 - Why is the Standard Model incompatible with the generation of gravitational waves coming from the primordial universe?
 - CFTC Summer Internship Program 2024 - A Tale of Two Vacua